

Habit Tracker App

Project Description

For our final project, we created a Habit Tracker App in Python that allows users to track habits over time in one place. Users can create an account, log in, save daily habit entries, view previous data, and generate graphs to visualize their progress. The app focuses on four habits: water consumption, reading, productivity, and sleep consistency.

Our goal with this project was to make something interactive, organized, and visually appealing while improving our coding skills in Python. We also wanted to practice working collaboratively through GitHub and debugging a larger project together.

Features

- User login and sign-up system
- Password hashing for account security
- Separate saved data files for each user
- Water consumption tracker
- Reading tracker
- Productivity tracker
- Sleep consistency tracker
- Habit summaries and averages
- Graphs showing progress over time
- Delete last entry feature
- Interactive user interface using ipywidgets

Libraries and Packages Used

We used the following Python libraries/packages:

- pandas
- matplotlib
- ipywidgets
- hashlib
- os
- datetime

How to Run the Project

1. Download all project files.
2. Install required packages using:

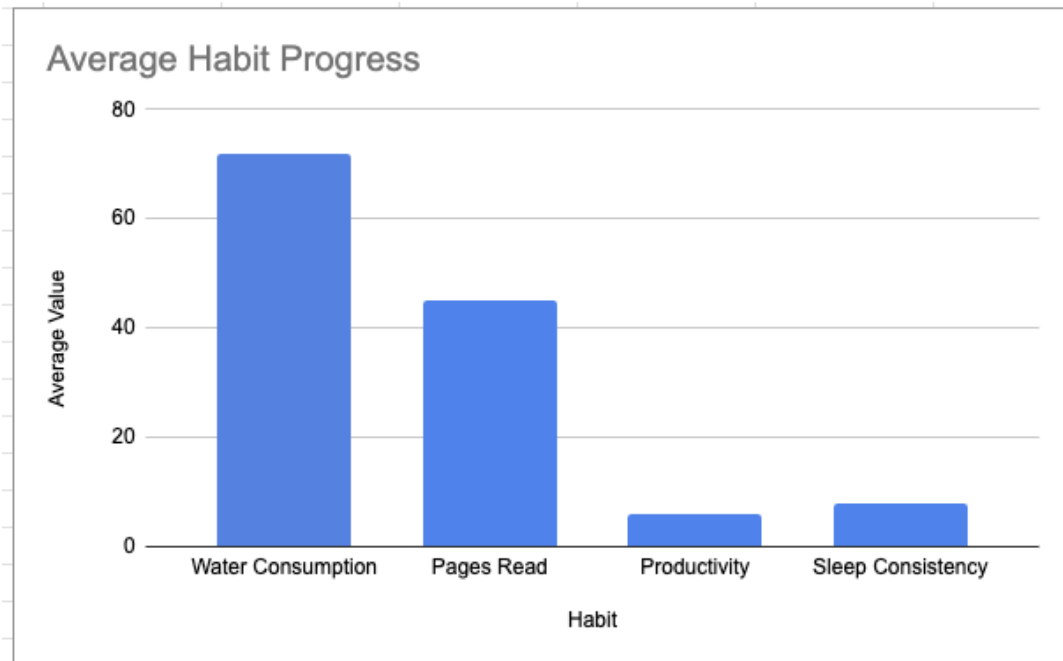
None

```
pip install pandas matplotlib ipywidgets
```

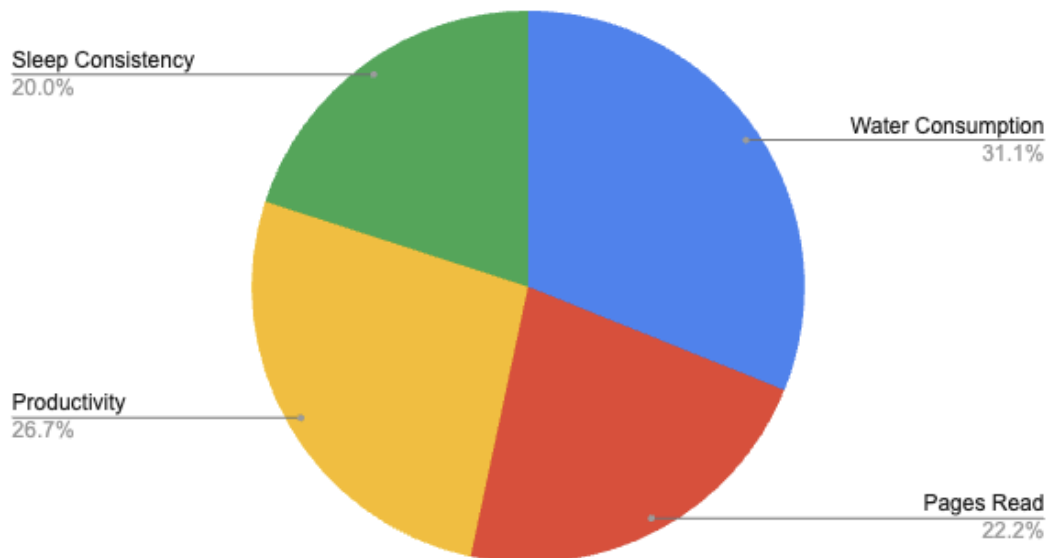
3. Run the Python file or notebook.
4. Create an account using the “Sign Up” button.
5. Log in and begin tracking habits.
6. Save entries, view summaries, and generate graphs using the buttons in the app.

Example Visualizations

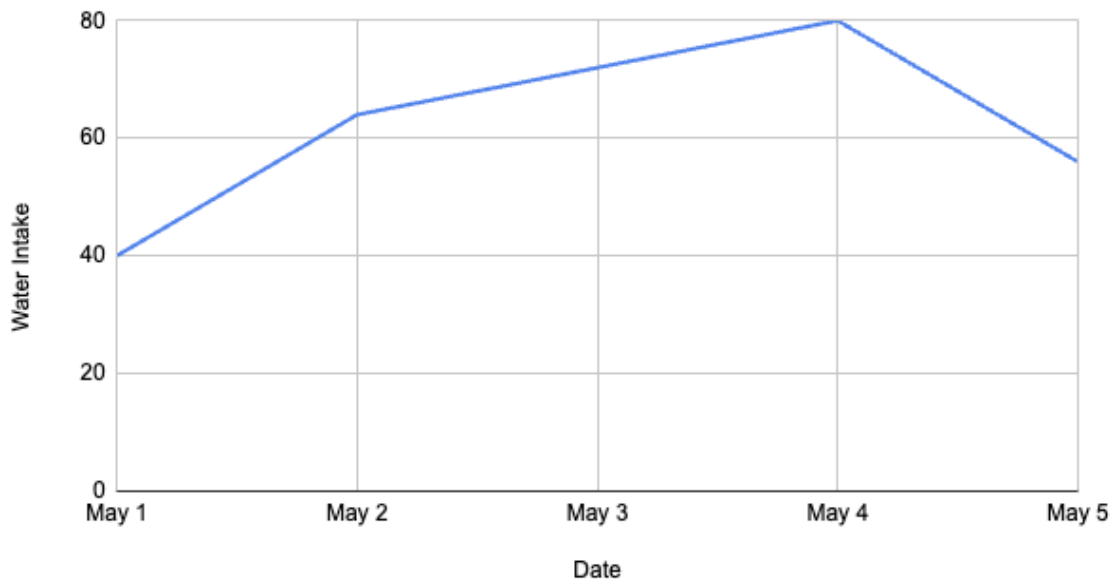
The app includes graphs that allow users to visualize their habit progress over time. These graphs help users compare habits and monitor consistency.



Number of Entries Per Habit



Water Intake Over Time



Team Contributions

Gianna Miller

Gianna worked on setting up the overall project structure, researching packages, creating separate user data files, implementing the login/sign-up system, building the habit input sections, and helping with testing and debugging.

Bowie Krawczyk

Bowie focused on improving the visual design and overall layout of the app, organizing the interface, improving graph readability, fixing formatting and indentation issues, testing functionality, and helping create the final presentation and README visuals.

Challenges and Reflection

One of the biggest challenges during this project was debugging indentation and formatting issues while trying to combine multiple interactive features into one application. We also spent a lot of time figuring out how to save and load separate user data correctly while keeping the app organized and easy to use.

Another challenge was improving the visual appearance of the app and graphs. We wanted the app to feel more polished and interactive rather than just displaying plain outputs.

Overall, this project helped us become more comfortable with Python, debugging, data visualization, and collaborative coding. It also showed us how important communication and organization are when working on larger coding projects with another person.

GitHub Information

Our repository includes regular commits from both group members, commented and organized code, and all files needed to run the project successfully.